Virtual Sensor & Thermal Modeling

An advanced, comprehensive breakdown of the physics, data pipelines, and AI behind TrustSense-BMS.

1. The Physics: How a Battery Gets Hot

> Before we use AI, we use pure math (PyBaMM) to calculate how much heat a battery is generating. AI must always be grounded in thermodynamic reality.

We calculate heat generation using three fundamental physics formulas:

Formula 1: Heat Generated (Joule Heating)

$$Q_{gen} = I^2 \times R(SOC, T)$$

Q_gen = The heat created inside the battery.

I = The Current.

R(SOC, T) = Internal Resistance. *(Note: Resistance is not static! It changes based on the battery's State of Charge (SOC) and current Temperature (T)).*

Formula 2: Heat Lost (Newton's Law of Cooling)

$$Q_{loss} = h \times A \times (T_{cell} - T_{coolant})$$

Q_loss = Heat removed by the car's cooling system.

h × A = Cooling efficiency and surface area.

T_cell - T_coolant = The difference in temperature between the battery and the coolant.

Formula 3: The Final Temperature Change (Lumped Thermal Mass)

$$m \cdot C_p \cdot \frac{dT}{dt} = Q_{gen} - Q_{loss}$$

m = Mass (Weight of the cell).

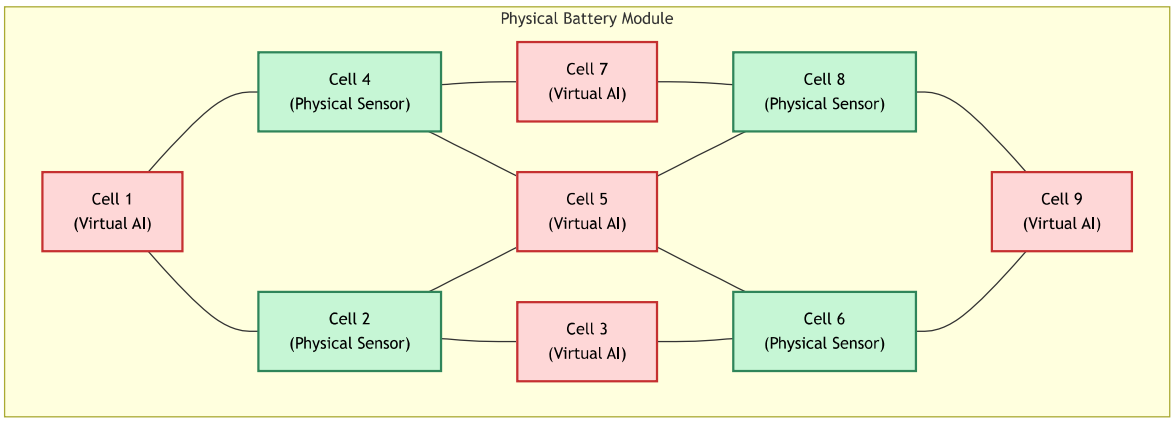
C_p = Specific Heat Capacity.

dT/dt = The rate the temperature is going up.

In plain English: (How heavy it is) × (Temperature change) = (Heat created) - (Heat removed).

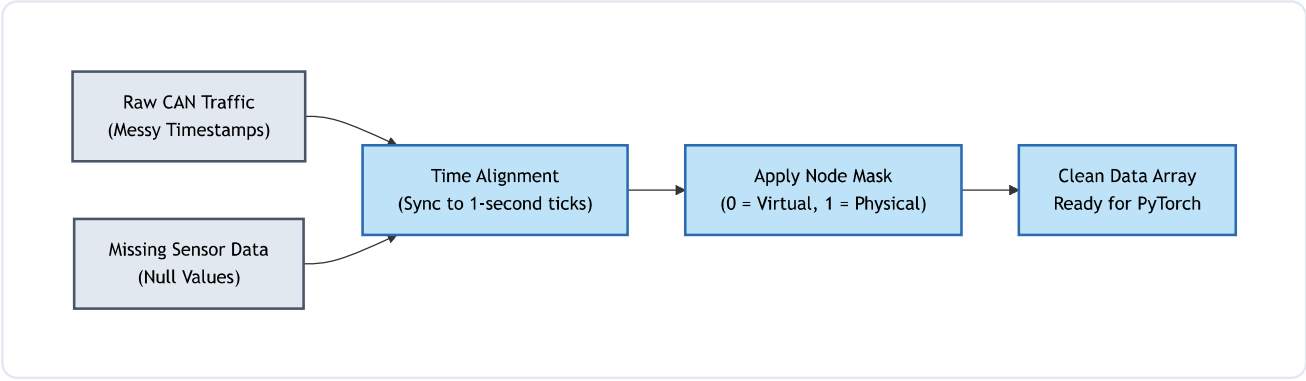
2. Visualizing the Battery Grid

Because we know the physics above, we don't need a thermometer on every single cell. We place physical sensors sparsely (in green). The AI handles the "Virtual" cells (in red).



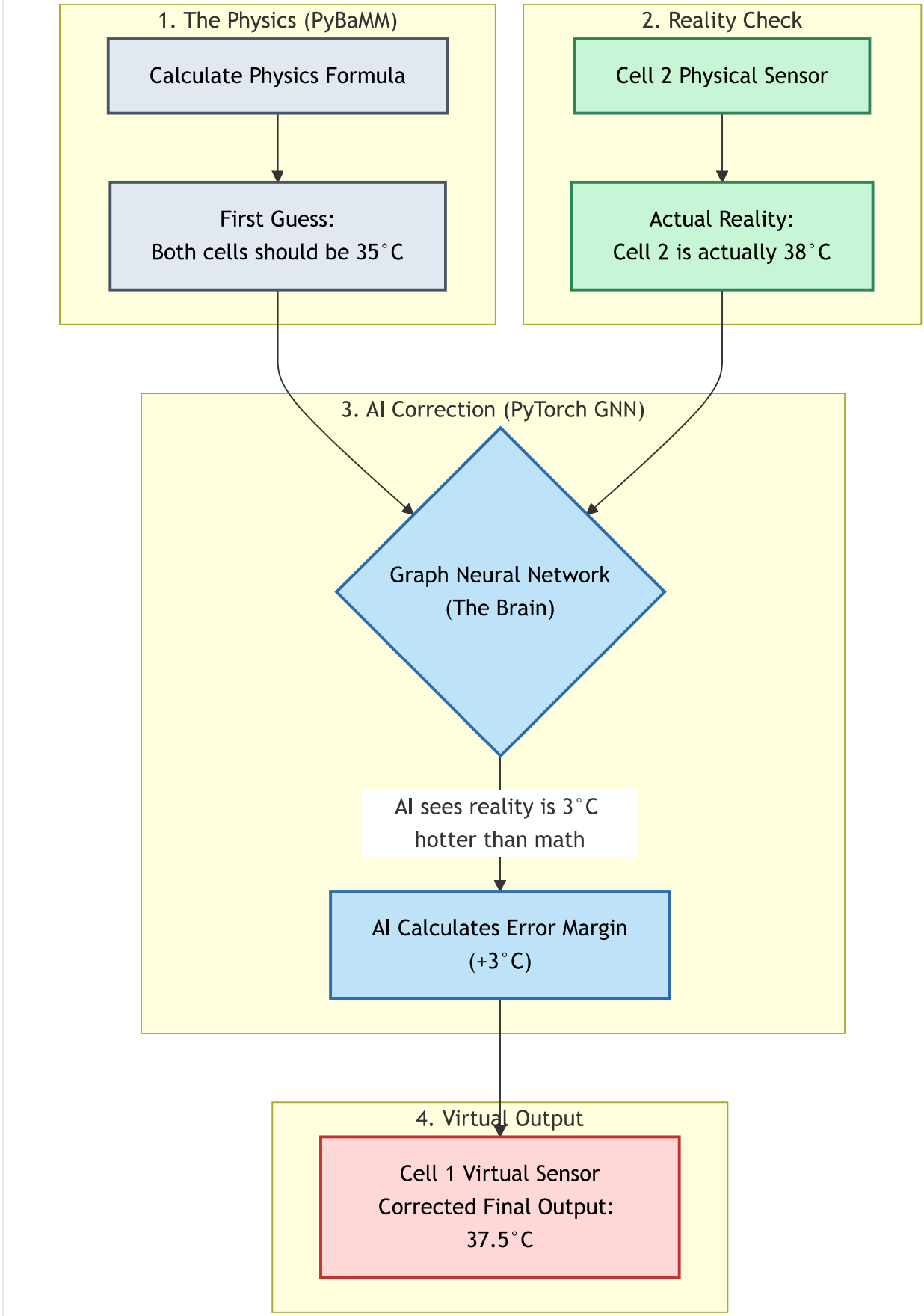
3. Data Pre-Processing Pipeline

> Raw CAN bus network data is messy. Messages arrive at different times, and virtual cells have missing temperature values (`null`). The data must be cleaned before the AI can see it.



4. How the AI (Graph Neural Network) Combines Math & Sensors

The AI looks at the difference between the **Math Prediction** and the **Real Sensor** to correct its final answer. Here is the exact logic pipeline:



5. The AI Math: Graph Message Passing

How does the AI actually update the temperatures? It uses a mathematical operation called **Message Passing**. Hot cells "pass a message" to their neighbors across the graph edges.

Graph Neural Network (Message Passing)

$$h_i^{(l+1)} = \sigma\left(\sum_{j \in N(i)} W \cdot h_j^{(l)}\right)$$

h_i = The temperature state of Cell 1.
j ∈ N(i) = Looking at all the neighboring cells (Cell 2, Cell 4, etc.).
W = The Neural Network's learned weight (Thermal resistance between cells).
In plain English: A cell updates its own temperature by looking at the temperatures of all the cells touching it, multiplied by how easily heat moves between them.

6. Sensor Ablation & Measuring Success

How do we know we can trust the AI? We remove sensors one-by-one and measure the error using the **RMSE (Root Mean Square Error)** formula. Our safety limit requires **RMSE ≤ 1.0°C**.

Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{T}_i - T_i)^2}$$

\hat{T}_i = The AI's guessed temperature.

T_i = The actual true temperature.

In plain English: We measure the gap between the guess and the truth, square it (so negatives don't cancel positives), average it out, and take the square root. If the final number is greater than 1.0, the model fails.

